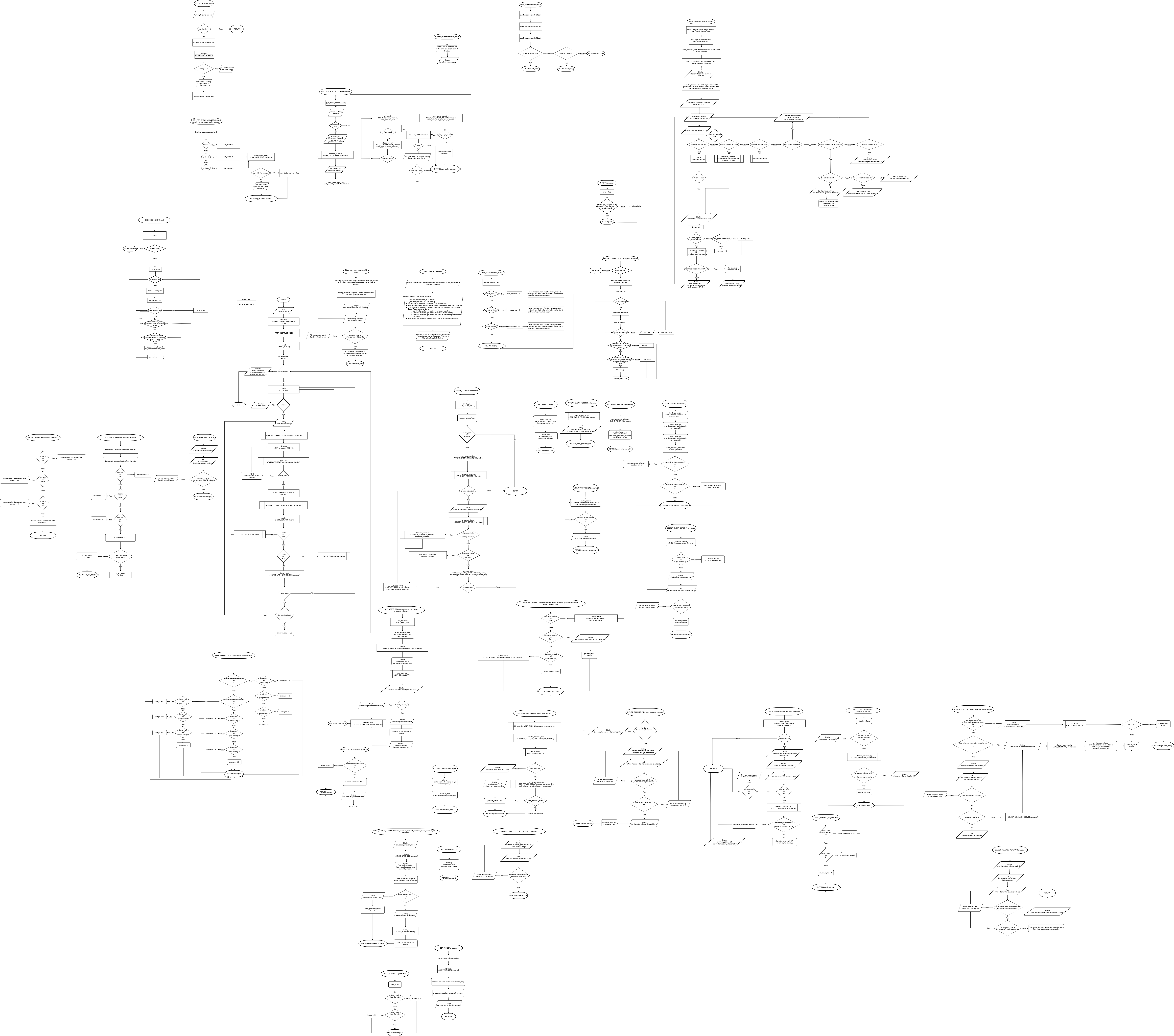
BUY_POTION(character)
Enter y to buy or n to skip
user_input = 'y'
False
RETURN
True
budget = money character has
change =
budget - POTION_PRICE
change >= 0
False
You can't buy with
your current budget
True
Purchase successful
Your change is
$(change).
money character has = change
CHECK_FOR_BADGE_CHANGE(character,
actual_win_count, gym_badge_earned)
level = character's current level
level == 1
True
win_count = 2
False
level == 2
True
win_count = 3
False
level == 3
True
win_count = 4
count_left_for_badge
= win_count - actual_win_count
count_left_for_badge > 0
False
gym_badge_earned = True
True
You need to win
{count_left_for_badge}
more time
RETURN(gym_badge_earned)
CHECK_LOCATION(board)
location = ""
board is empty
True
RETURN(location)
False
row_index = 0
row_index < rows
Text
Create an empty row
column_index = 0
column_index < rows
False
row_index += 1
True
coordinate of row_index
and column_index exists on the
board
False
True
coordinate of row_index
and column_index == character's
current location
True
location = coordinate of
row_index and column_index
column_index += 1
CONSTANT
POTION_PRICE = 10
START
Ask
character name
character
= MAKE_CHARACTER(character
input)
PRINT_INSTRUCTIONS()
board
= MAKE_BOARD()
achieved_goal
= False
achieved_goal
True
Display
"Congratulations!
You have successfully
finished your journey :)"
False
check
= IS_ALIVE()
check
False
Display
"Game Over"
END
True
Display
current character status
DISPLAY_CURRENT_LOCATION(board, character)
direction
= GET_character_CHOICE()
valid_move
= VALIDATE_MOVE(board, character, direction)
valid_move
False
Display
character can't go the
direction
True
MOVE_CHARACTER(character,
direction)
DISPLAY_CURRENT_LOCATION(board, character)
location
= CHECK_LOCATION(board)
location
==
'store'
True
BUY_POTION(character)
False
location
==
'gym'
False
EVENT_OCCURRED(character)
True
battle_result
= BATTLE_WITH_GYM_LEADER(character)
battle_result
False
True
character level <= 3
True
False
achieved_goal = True
MAKE_CHARACTER(character
name)
character_status contains data about money, poke ball, current
level, potion, current location, character name, starting
pokemon
starting_pokemon = Squirtle, Charmander, Bulbasaur
with their type and currentHP
Display
starting pokémon list with their type
Ask
which starting pokémon
the character wants
character input is
in the starting pokémon list
False
Tell the character about
that it is not valid option
True
Put character input pokémon
into poké ball with its type and HP
and starting pokémon
RETURN(character_status)
BATTLE_WITH_GYM_LEADER(character)
gym_badge_earned = False
Enter y to challenge,
n to quit
character_input
== 'y'
False
True
Gym leader:
Welcome to the gym!
Here is one rule,
you can't use potions
selected_pokemon
= TAKE_OUT_POKEMON(character)
You have chosen
{selected_pokemon}
gym_leader_pokemon =
GET_EVENT_POKEMON(character)
fight_result =
FIGHT(character_pokemon,
event_pokemon_info)
fight_result
False
True
attacked_result
= GET_ATTACKED(event_pokemon,
event_type, character_pokemon)
attacked_result
True
gym_badge_earned =
CHECK_FOR_BADGE_CHANGE(character,
actual_win_count, gym_badge_earned)
gym_badge_earned
False
alive = IS_ALIVE(character)
alive
False
True
Enter y if you want to proceed another
battle in the gym, else n
user_input == 'y'
False
True
True
character's current
level += 1
RETURN(gym_badge_earned)
describe_location(character_status)
Find the cell on the board that
matches the character's current
location
Display
character's current location
make_board(character_status)
level1_map represents 26 cells
level2_map represents 32 cells
level3_map represents 42 cells
character's level == 1
False
character's level == 2
False
RETURN(level3_map)
True
RETURN(level1_map)
True
RETURN(level2_map)
IS_ALIVE(character)
alive = True
At least one Pokemon in the
character's Poke Ball has HP
greater than 0
False
alive = False
True
RETURN(alive)
event_happened(character_status)
event_collection contains wildPokemon,
teamRocket, strangeTrainer
event_type is a random event
from event_collection
event_pokemon_collection contains data about 20kinds
of wild pokemon
event_pokemon is a random pokemon from
event_pokemon_collection
Display
what event pokemon shows up
with HP
character_pokemon is a random pokemon with HP
greater than 0 that will face the event Pokémon from
the poke ball from character_status
Display the character's Pokémon
along with its HP
Display what options
the character can choose
Ask what the character wants to do
character_input ==
"fight"
character choose "fight"
False
character choose "Pokemon"
False
character choose "Potion"
False
event_type is wildPokemon
True
result
= fight(character_status)
True
character_pokemon =
change_pokemon(character_status,
character_pokemon)
True
potion(character_status)
result == True
True
False
Let the character know
it is wrong choice,
and rechoose correct option
True
character choose "Throw Poke Ball"
False
character choose "Run"
False
True
Display
character ran away
from the wild pokemon successfully
True
the wild pokemon's HP < 5
False
the wild pokemon broke free
True
Let the character know
the wild pokemon broke free
True
Let the character know
the character caught the wild pokemon
False
Let the character know
the character failed to get the wild pokemon
Add the wild pokemon to the
poke ball in the
character_status
Display
which skill the event pokemon used
damage = 1
event_type is
wildPokemon
False
event_type is teamRocket
True
damage = 1.5
False
damage = 1.3
True
the character pokemon
HP
-= skilldamage * damage
the character pokemon's HP <= 0
True
the character
pokemon's HP = 0
Let the character know
character's pokemon fainted
False
Display
how many damage
the character's pokemon got
and how many HP left
PRINT_INSTRUCTIONS()
Welcome to the world of Pokemon! Embark on an exciting journey to become a
Pokemon Champion.
Important notes to know before you begin:
Stores are represented by S on the map.
Gyms are represented by G on the map.
If all six of your Pokémon lose their HP, the game is over.
You can only challenge a gym leader once you have a full team of six Pokémon.
After defeating a gym leader, you will earn a badge, unlocking the next level.
Badge Requirements by Level:
Level 1: Defeat the gym leader twice to earn a badge.
Level 2: Defeat the gym leader three times to earn a badge.
Level 3: Defeat the gym leader four times to earn a badge and complete the mission.
The mission is complete when you defeat the final Gym Leader at Level 3.
Your journey will be tough, but with determination
and strategy, you can become a Pokémon
Champion. Good luck, Trainer!
RETURN
MAKE_BOARD(current_level)
Create an empty board
current_level == 1
True
rows, columns = 6, 6
Inside the board, mark True for the playable 4x6
rectangle and the 2 extra cells for the start and end,
and mark False for all other cells
False
current_level == 2
True
rows, columns = 5, 8
Inside the board, mark True for the playable 5x6
rectangle and the 2 extra cells for the start and end,
and mark False for all other cells
False
current_level == 3
True
rows, columns = 5, 10
Inside the board, mark True for the playable 5x8
rectangle and the 2 extra cells for the start and end,
and mark False for all other cells
False
RETURN(board)
DISPLAY_CURRENT_LOCATION(board, character)
board is empty
True
RETURN
Find the maximum row and
column in the board
row_index = 0
row_index < rows
Text
Create an empty row
column_index = 0
column_index < rows
False
Print row
row_index += 1
True
coordinate of row_index
and column_index exists on the
board
False
row += "   "
True
coordinate of row_index
and column_index == character's
current location
False
row += "[ ]"
True
row += "[#]"
column_index += 1
MOVE_CHARACTER(character, direction)
direction
==
up
True
current location Y-coordinate from
charater -= 1
False
direction
==
down
True
current location Y-coordinate from
charater += 1
False
direction
==
left
True
current location X-coordinate from
charater -= 1
False
current location X-coordinate from
charater += 1
RETURN
VALIDATE_MOVE(board, character, direction)
Y-coordinate = current location from character
X-coordinate = current location from character
direction
==
up
True
Y-coordinate -= 1
False
direction
==
down
True
Y-coordinate += 1
False
direction
==
left
True
X-coordinate -= 1
False
X-coordinate += 1
X-, Y-coordinate are
in the board
False
on_the_board
= False
True
on_the_board
= True
RETURN(on_the_board)
GET_CHARACTER_CHOICE()
Display
a numbered list of directions
Ask
which direction
the character wants to choose
character input is
in a numbered list of directions
False
Tell the character about
that it is not valid option
True
RETURN(character input)
EVENT_OCCURRED(character)
event_type
= SET_EVENT_TYPE()
process_result = True
event_type
==
No event
True
False
event_pokemon_info
= APPEAR_EVENT_POKEMON(character)
character_pokemon
= TAKE_OUT_POKEMON(character)
process_result
False
RETURN
True
Display
what the character's pokemon is with HP
character_choice
= SELECT_EVENT_OPTION(event_type)
character_choice
==
change pokemon
True
character_pokemon
= CHANGE_POKEMON(character,
character_pokemon)
False
character_choice
==
use potion
True
USE_POTION(character,
character_pokemon)
False
process_result
= PROCEED_EVENT_OPTION(character_choice,
character_pokemon, character, event_pokemon_info)
process_result
True
process_result
= GET_ATTACKED(event_pokemon,
event_type, character_pokemon)
False
SET_EVENT_TYPE()
event_collection
= Wild pokemon, Team Rocket,
Strange trainer, No event
event_type
= a random event
from event_collection
RETURN(event_type)
APPEAR_EVENT_POKEMON(character)
event_pokemon_info
= GET_EVENT_POKEMON(character)
Display
what type of event occurred
and what event pokemon is with its HP
RETURN(event_pokemon_info)
GET_EVENT_POKEMON(character)
event_pokemon_collection
= EVENT_POKEMON(character)
event_pokemon_info
= a random pokemon
from event_pokemon_collection
with its type and HP
RETURN(event_pokemon_info)
EVENT_POKEMON(character)
level1_pokemon
= level1 pokemon collection with
their type and HP
level2_pokemon
= level2 pokemon collection with
their type and HP
level3_pokemon
= level3 pokemon collection with
their type and HP
event_pokemon_collection
= level1_pokemon
current level (from character)
==
2
True
event_pokemon_collection
= level2_pokemon
False
current level (from character)
==
3
True
event_pokemon_collection
= level3_pokemon
False
RETURN(event_pokemon_collection)
TAKE_OUT_POKEMON(character)
character_pokemon
= a random pokemon with its type and HP
from poke ball (from character)
character_pokemon's HP
==
0
True
False
Display
what the character pokemon is
RETURN(character_pokemon)
SELECT_EVENT_OPTION(event_type)
character_option
= Fight, Change pokemon, Use potion
event_type
==
Wild pokemon
True
character_option
+= Throw poke ball, Run
False
Display
what options the character has
Ask
what option the character wants to choose
character input is included
in character_option
False
Tell the character about
that it is not valid option
character_choice
= character input
RETURN(character_choice)
PROCEED_EVENT_OPTION(character_choice, character_pokemon, character,
event_pokemon_info)
character_choose
==
fight
True
process_result
= FIGHT(character_pokemon,
event_pokemon_info)
False
character_choose
==
Run
True
Display
the character escaped from event pokemon
process_result
= False
False
character_choose
==
Throw poke ball
True
process_result
= THROW_POKE_BALL(event_pokemon_info, character)
False
process_result = False
RETURN(process_result)
MAKE_DAMAGE_STRONGER(event_type, character)
current level(from character)
==
1
True
event_type
==
team rocket
True
stronger = 1.4
False
event_type
==
strange trainer
True
stronger = 1.3
False
event_type
==
wild pokemon
True
stronger = 1
False
stronger = 1.5
False
current level(from character)
==
2
True
event_type
==
team rocket
True
stronger = 1.7
False
event_type
==
strange trainer
True
stronger = 1.5
False
event_type
==
wild pokemon
True
stronger = 1.3
False
stronger = 2
False
event_type
==
team rocket
True
stronger = 1.9
False
event_type
==
strange trainer
True
stronger = 1.7
False
event_type
==
wild pokemon
True
stronger = 1.5
False
stronger = 2.5
RETURN(stronger)
GET_ATTACKED(event_pokemon, event_type,
character_pokemon)
skill_collection
= GET_SKILL_OF()
event_pokemon_skill
= a random skill from the
skill_collection
damage
= MAKE_DAMAGE_STRONGER(event_type, character)
damage
*= a random number
from the skill damage range
skill_accuracy
= GET_PROBABILITY()
Display
what kind of skill the event pokemon used
skill_accuracy
False
Display
the event pokemon's skill missed
True
Display
the event pokemon's skill hit
character_pokemon's HP -=
damage
Display
how many damage
character_pokemon got
process_result
= CHECK_STATUS(character_pokemon)
RETURN(process_result)
CHECK_STATUS(character_pokemon)
character pokemon's HP
<=
0
False
status = True
RETURN(status)
True
character pokemon's HP = 0
Display
the character pokemon fainted
status = False
GET_SKILL_OF(pokemon_type)
skills_of
= skill collection depending on type
with damage range
pokemon_skill
= skill collection of pokemon_type
RETURN(pokemon_skill)
FIGHT(character_pokemon, event_pokemon_info)
skill_collection = GET_SKILL_OF(character_pokemon's type)
character_pokemon_skill
= CHOOSE_SKILL_TO_CHALLENGE(skill_collection)
skill_accuracy
= GET_PROBABILITY()
skill_accuracy
False
Display
character_pokemon's skill missed
Display
event pokemon's HP
(from event_pokemon_info)
True
event_pokemon_status
= GET_ATTACK_RESULT(character_pokemon_skill,
skill_collection, event_pokemon_info, character)
event_pokemon_status
True
process_result = True
False
process_result = False
RETURN(process_result)
CHANGE_POKEMON(character, character_pokemon)
Number of
the character's Pokémon
==
1
True
Display
the character has no pokemon to switch to
False
Display
the character pokemons' status
from poke ball (from character)
Ask
Which Pokémon the character wants to switch to
character input is included
in the character pokemon list
False
Tell the character about
that it is not valid option
True
character input pokemon' HP
==
0
True
Tell the character about
the pokemon has 0 HP
False
Display
the character pokemon is switching to
character_pokemon
= character input
RETURN(character_pokemon)
USE_POTION(character, character_pokemon)
validate_potion
= CHECK_POTION(character,
character_pokemon)
validate_potion
False
RETURN
True
Display
how many potions the character has
(from character)
Display
character_pokemon's status
Ask
the character wants to use a potion
Tell the character about
that it is not valid option
False
character input
==
no
True
character input
==
yes
False
True
pokemon_maximum_hp
= LEVEL_MAXIMUM_HP(character)
character_pokemon's HP
>=
pokemon_maximum_hp - 5
False
character_pokemon's HP += 5
True
character_pokemon's HP
= pokemon_maximum_hp
Display
how many potions left
and what character_pokemon's HP
CHECK_POTION(character,
character_pokemon)
validation = False
the amount of potion
the character has
==
0
True
Display
the character has no potion
False
pokemon_maximum_hp
= LEVEL_MAXIMUM_HP(character)
character_pokemon's HP
==
pokemon_maximum_hp
True
Display
character_pokemon has full HP
False
validation = True
RETURN(validation)
THROW_POKE_BALL(event_pokemon_info, character)
event pokemon's HP
(from event_pokemon_info)
<=
5
False
Display
the character failed
to catch the event pokemon
run_or_not
= GET_PROBABILITY()
run_or_not
True
process_result
= True
False
True
Total pokemon number the character has
==
6
False
Display
what pokemon the character caught
pokemon_maximum_hp
= LEVEL_MAXIMUM_HP(character)
Add the event pokemon
to the character pokemon collection
with its type and a half of
pokemon_maximum_hp
process_result
= False
RETURN(process_result)
True
Display
the character has up to 6 pokemons
Ask
the character wants to release
one character pokemon
character input is yes or no
False
Tell the character about
that it is not valid option
True
character input is no
False
SELECT_RELEASE_POKEMON(character)
True
Tell
the event pokemon broke free
GET_ATTACK_RESULT(character_pokemon_skill, skill_collection, event_pokemon_info,
character)
Display
character_pokemon_skill hit
damage
= MAKE_STRONGER(character)
damage
*= a random number
from the skill damage range
from skill_collection
event pokemon's HP (from
event_pokemon_info) -= damage
event pokemon's HP
<=
0
False
Display
event pokemon's HP, name
event_pokemon_status
= True
RETURN(event_pokemon_status)
True
Display
event pokemon is defeated
money
= GET_MONEY(character)
event_pokemon_status
= False
GET_PROBABILITY()
accuracy
= a random result
between True or False
RETURN(accuracy)
CHOOSE_SKILL_TO_CHALLENGE(skill_collection)
Display
what skills character_pokemon can use
with damage range
Ask
what skill the character wants to use
character input is included
inside character_option
False
Tell the character about
that it is not valid option
True
RETURN(character input)
LEVEL_MAXIMUM_HP(character)
current level
(from character)
==
1
True
maximum_hp = 30
False
current level
==
2
True
maximum_hp = 50
False
maximum_hp = 80
RETURN(maximum_hp)
SELECT_RELEASE_POKEMON(character)
Display
all of character's pokemons with HP
Tell
the character can't choose
starting pokemon
Ask
what pokemon the character release
The character input is included in the
character's Pokémon collection
False
Tell the character about
that it is not valid option
True
The character input is
the character's starting pokemon
True
False
Remove the character input pokemon's information
from the character pokemon collection
Display
the character released character input pokemon
RETURN
GET_MONEY(character)
money_range = three numbers
money =
MAKE_STRONGER(character)
money *= a random number from money_range
character money(from character) += money
Display
how much money the character got
RETURN
MAKE_STRONGER(character)
stronger = 1
current level
(from character)
==
2
True
stronger = 1.3
False
current level
(from character)
==
2
True
stronger = 1.3
False
RETURN(stronger)